



# Project

## MIRROR

An AI Powered Memory and Emotional Analytics System

Presented by:

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Project Guide:

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# Outline

1. Data Collection Progress
2. Analysis & Interpretation (Preliminary)
3. Consistency with Objectives



# Data Collection Progress

## **Authenticity of Data:**

Mirror Captures real time user reflections through our frontend using two natural input methods:

1. Direct text typing
2. Built in speech to text processing (Voice recording)



Good evening,  
**Prabhu Teja Pamula**

*"Who looks outside, dreams; who looks inside, awakes."*

CARL JUNG



hey there





Good evening,  
**Prabhu Teja Pamula**

*"Who looks outside, dreams; who looks inside, awakes."*

CARL JUNG



Listening & transcribing...





# Data Collection Progress

## Adequacy of Data Collected:

Ensures no information is lost by storing all 7 key details needed for the AI to work properly:

**id:** Unique database identifier.

**content:** The actual journal text.

**created\_at:** The exact time the entry was saved.

**emotion:** The dynamically generated mood tag.

**sender:** Identifies who spoke (the user or mirror).

**user\_id:** Securely links the entry to the logged in user.

**embedding:** The mathematical vector used by the AI to find patterns.



| <div><div><div><div></div></div></div><div><div>DATA</div><div>STRUCTURE</div></div><div><div>&lt;</div><div>&gt;</div></div><div><div>Filters</div></div><div><div>Sort</div></div><div><div>Columns</div></div><div><div>+ Add record</div></div></div> <div>817ms</div> <div><div>&lt;</div><div>1 - 50 of 50+</div><div>&gt;</div></div> |              |                           |                            |                           |                    |                      |                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------|----------------------------|---------------------------|--------------------|----------------------|---------------------------|
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                     | id bigserial | content text              | created_at timestamp wi... | emotion varchar(255)      | sender varchar(50) | user_id varchar(255) | embedding vector(768)     |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                     | 1            | Hey                       | 2026-06-03 20:31:30.89...  | COGNITION Warm Opening... | user               | Prabhu Teja Pamula   | [-0.024764938,0.019171... |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                     | 2            | Hey there! It's great ... | 2026-06-03 20:31:31.18...  | COGNITION Warm Opening... | mirror             | Prabhu Teja Pamula   | NULL                      |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                     | 3            | Na chilling exploring ... | 2026-06-03 20:32:15.71...  | COGNITION Peaceful Exp... | user               | Prabhu Teja Pamula   | [0.010331861,0.0092440... |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                     | 4            | Oh, that sounds like a... | 2026-06-03 20:32:15.97...  | COGNITION Peaceful Exp... | mirror             | Prabhu Teja Pamula   | NULL                      |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                     | 5            | True music is beautifu... | 2026-06-03 20:39:08.90...  | FEELINGS Profound Musi... | user               | Prabhu Teja Pamula   | [-0.03619591,-0.010454... |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                     | 6            | That's such a lovely t... | 2026-06-03 20:39:09.12...  | FEELINGS Profound Musi... | mirror             | Prabhu Teja Pamula   | NULL                      |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                     | 7            | Damm this mirror app f... | 2026-06-03 20:42:10.56...  | COGNITION Intrigued Ap... | user               | Prabhu Teja Pamula   | [-0.007342496,-0.01386... |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                     | 8            | That's so wonderful to... | 2026-06-03 20:42:10.82...  | COGNITION Intrigued Ap... | mirror             | Prabhu Teja Pamula   | NULL                      |
| <input type="checkbox"/>                                                                                                                                                                                                                                                                                                                     | 9            | 🔥                         | 2026-06-03 20:43:40.46...  | FEELINGS Joyful Groove... | user               | Prabhu Teja Pamula   | [-0.019391904,0.000094... |



# Data Collection Progress

## **Proper Tools Used (Relational Architecture):**

### **Neon Serverless PostgreSQL:**

A modern cloud database that securely manages user accounts and active sessions.

### **pgvector Extension:**

A powerful tool that allows us to store the AI's mathematical vectors natively inside our SQL tables.



Prabhu Teja

Free

mirror-db

All OK

Ask AI



Upgrade



PROJECT

- Dashboard
- Branches
- Integrations
- Settings

BRANCH

production

- Overview
- Monitoring
- SQL Editor
- Tables
- Backup & Restore
- Data Masking BETA

APP BACKEND

- Data API
- Auth
- Collapse menu

Tables

production

neondb

Schema

public

Search...

- auth\_flyway\_schema\_history
- memories
- memory\_flyway\_schema\_history
- otp\_tokens
- refresh\_tokens
- users



+ Add record

450ms



1 - 50 of 50+



ening...

user

Prabhu Teja Pamula

[-0.024764938,0.019171...



ening...

mirror

Prabhu Teja Pamula

NULL



l Exp...

user

Prabhu Teja Pamula

[0.010331861,0.0092440...



l Exp...

mirror

Prabhu Teja Pamula

NULL



Musi...

user

Prabhu Teja Pamula

[-0.03619591,-0.010454...



Musi...

mirror

Prabhu Teja Pamula

NULL



ed Ap...

user

Prabhu Teja Pamula

[-0.007342496,-0.01386...



ed Ap...

mirror

Prabhu Teja Pamula

NULL



roove...

user

Prabhu Teja Pamula

[-0.019391904,0.000094...



roove...

mirror

Prabhu Teja Pamula

NULL



Admi...

user

Prabhu Teja Pamula

[-0.019166764,0.005411...



Admi...

mirror

Prabhu Teja Pamula

NULL



ck Ap...

user

Prabhu Teja Pamula

[-0.028929086,0.002112...



ck Ap...

mirror

Prabhu Teja Pamula

NULL



ntial...

user

Prabhu Teja Pamula

[0.0050910222,-0.00002...



ntial...

mirror

Prabhu Teja Pamula

NULL



t Cre...

user

Prabhu Teja Pamula

[0.0017602629,0.005602...



# Analysis & Interpretation (Preliminary)

## **Use of Appropriate Analytical Tools:**

## **Microservices Architecture:**

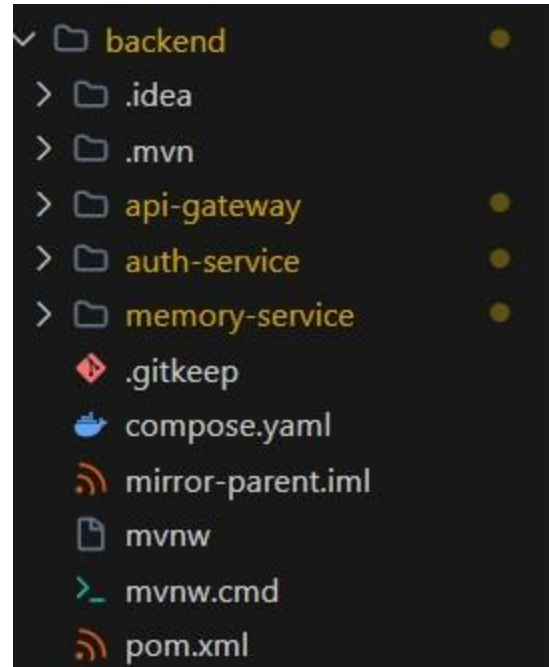
The backend is split into independent services (api-gateway, auth-service, memory-service) to process user data rapidly and keep the code organized.

## **Asynchronous Processing:**

Uses RabbitMQ to handle heavy AI and emotion tagging tasks in the background so the app never freezes while processing text.

## **High Speed Caching:**

Uses an Upstash Redis cache database to store computed text embeddings instantly, ensuring lightning fast search responses.



Azure services



Create a resource



Cost Management ...



App Services



Quickstart Center



Foundry



Kubernetes services



Virtual machines



Storage accounts



SQL databases



More services

Resources

Recent Favorite

| Name                                                                                                     | Type             | Last Viewed       |
|----------------------------------------------------------------------------------------------------------|------------------|-------------------|
|  miror-memory-service   | App Service      | a few seconds ago |
|  miror-zipkin           | App Service      | 2 days ago        |
|  miror-api-gateway      | App Service      | 6 days ago        |
|  miror-auth-service   | App Service      | 6 days ago        |
|  ASP-mirrorgroup-bca0 | App Service plan | 3 weeks ago       |
|  miror-group          | Resource group   | 3 weeks ago       |

See all



## CLUSTER



Overview



Nodes



Logs

## CLIENTS



Connections



Channels



Consumers

## MESSAGING



Exchanges



Queues

## BRIDGES



Shovels



Federation

## ADMINISTRATION



Virtual hosts



Policies



Operator policies



Users

## Overview

1

[Connections](#)

3

[Channels](#)

2

[Consumers](#)

7

[Exchanges](#)

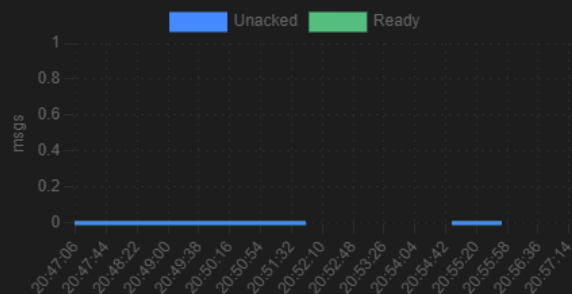
2

[Queues](#)

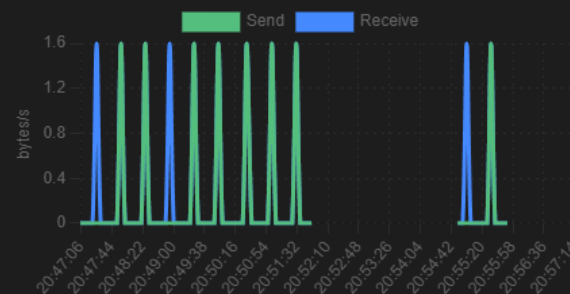
4d, 13h, 30m

Uptime

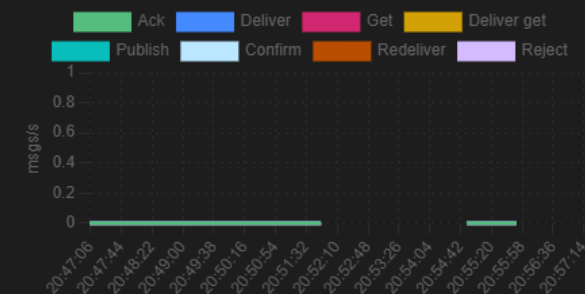
## Queued messages



## Data rates



## Message rates



## Upload definitions

Virtual host

All



File

Choose File

No file chosen

Upload



## Export definitions

Virtual host

All



Download



+

Search

+ Key

&gt;

99 embeddings::Sooo what to do with my life

TTL: No ▾

Text ▾

[illegible]

Save





# Analysis & Interpretation (Preliminary)

## **Logical Interpretations:**

## **Smart Emotion Tagging:**

Mirror doesn't rely on a fixed list of basic emotions. Instead, it analyzes the entry and instantly creates a custom emotion tag that captures the user's exact mood.

## **Interactive Visualizer:**

Takes the complex AI data from the backend and turns it into easy to read, dynamic graphs right on the user's screen.

## **Finding Past Patterns:**

Logically connects older journal entries to show clear trends in how the user's emotions and mindset are developing over time.



It seems the curiosity is still there, and I'm happy to engage in a conversation with you, just like we have before. What's on your mind, and how can I help you explore your thoughts?



CURIOUS ENGAGEMENT

8:37 PM

what ya doing

8:37 PM



The universe's vastness is a mystery that has captivated human imagination for centuries, with estimates suggesting it contains over 200 billion galaxies, each containing billions of stars, a scale that's almost incomprehensible to our everyday perception, yet it sparks a deep sense of cosmic wonder and awe within us



COSMIC WONDER

8:38 PM

how big is the universe?

8:38 PM



Speak or type...





PRABHU TEJA

API Error

Warm Curiosity

Heartfelt Creative Pride

Curious Engagement





# Consistency with Objectives

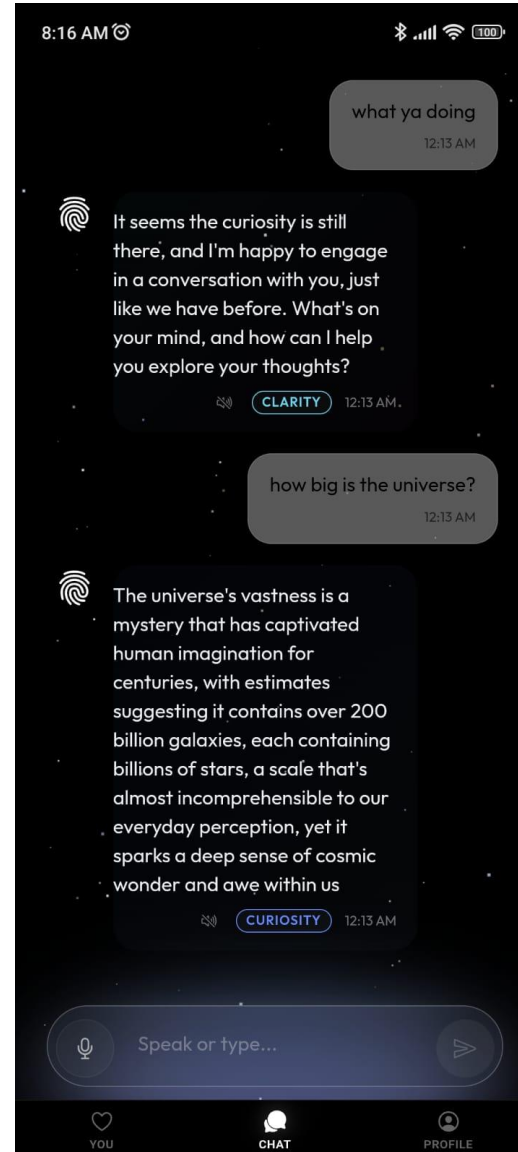
## **Work Progress Aligned with Original Objectives:**

### **One App for All Platforms:**

Successfully built the core screens (Login, Chat, You Profile) to work perfectly and look consistent on both Web and Android.

### **Strict Privacy Focus:**

The database and secure login (JWT) are designed to keep user data completely private. The project stays true to its original goal of being a purely *personal* journal, avoiding any public social features.





# Consistency with Objectives

## **Adherence to Scope:**

## **Low Friction User Experience:**

Built a clean, intuitive interface using Ionic and Angular, allowing users to effortlessly log their daily reflections through both text and voice inputs.

## **Cloud Native Distributed Architecture:**

The application stack is fully deployed across a distributed cloud environment. The core microservices and Zipkin are hosted on Azure, while Redis is hosted on Upstash and RabbitMQ on CloudAMQP, proving the system is a highly scalable, production ready product.



# Thank You

Questions and Feedback

**Repository:** <https://github.com/prabhu-tejaa/MIRROR>  
**Web Interface:** <https://projectmirror.tech/>  
**Android Build:** <https://github.com/prabhu-tejaa/MIRROR/releases>

MIRROR

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